Chapter 18

Minimal preparation (resin retained) bridges

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WHAT ARE MINIMAL PREPARATION (RESIN RETAINED) BRIDGES AND WHEN ARE THEY USED?

Minimal preparation bridges have also been referred to in the literature as resin retained bridges, adhesive bridges and resin-bonded bridges. In this text the term minimal preparation (resin retained) bridge will be used. These are bridges that require minimal preparation of tooth tissue and rely on adhesive cements for their primary retention, unlike conventional bridges that require significant tooth reduction to make retentively shaped preparations which provide the primary retention and then rely on a cement lute for secondary retention.

Minimal preparation (resin retained) bridges are therefore used when the potential abutment teeth are unrestored or minimally restored and there is a need to reduce unnecessary tooth preparation. Consider the 22-year-old patient in Figure 18.1: preparation of the vital upper right central incisor tooth for a metal–ceramic retainer has been destructive of tooth tissue as this tooth only had a small fractured incisal edge. A cantilevered minimal preparation (resin retained) bridge using the central incisor as an abutment tooth would have been much more conservative. However, in this patient both lateral incisors are missing. Conventional bridge retainer preparation of the non-vital upper left central incisor tooth has been necessary not only to retain a pontic but also to disguise the appearance of the discoloured tooth. Furthermore, this patient had orthodontic treatment to close a midline diastema and since this treatment tends to be unstable, the conventional retainers were linked to also act as an orthodontic retainer.

HISTORY OF RESIN-BONDED BRIDGES

It is useful to understand the history of the development of minimal preparation (resin-bonded) bridges as some principles and applications can be useful in contemporary practice.

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Figure 18.1 This is the patient seen in Figure 19.2, Chapter 19. Preparation of the upper right central incisor for a metal–ceramic retainer has not been conservative of tooth tissue.

Macromechanical retention

Natural tooth pontic

This technique can be used for a tooth that needs to be extracted. Once extracted, the root is resected and then the crown is cemented to the adjacent tooth/teeth using composite resin cement. At the resected root end the root canal should be opened and restored with either a glass ionomer or composite resin to prevent food and plaque stagnating within the tooth. To bond the extracted tooth the corresponding approximal surfaces are acid etched, bond applied and then composite used to cement the crown in place. Figure 18.2 illustrates diagrammatically how this procedure is carried out and Figure 18.3 shows an example where this has been done. In the author's opinion this technique has been remarkably successful for the patient in Figure 18.3 as this 'temporary' measure had been carried out many years previously.

If a patient presents with a missing tooth, which is either lost or not reusable, a replacement tooth can be made from composite resin using a crown former or simply an appropriately sized acrylic denture tooth can be used. Additional retention can be obtained, if required, by embedding a contoured orthodontic wire, wire mesh or fibre-reinforced ribbon in composite to the lingual surface of the extracted tooth and the tooth or teeth to which it is being bonded.

Laboratory-made bridges using macromechanical retention

Rochette bridges

A Rochette bridge consists of a pontic connected to a metal wing retainer which covers the lingual surface of the abutment tooth. Within the wing, countersunk holes are cut which allow retentive composite plugs or rivets to form when the bridge is cemented in situ (Figure 18.4). When Rochette bridges were originally described, they were designed to act as permanent fixed prosthodontic restorations (Figure 18.5). However, to achieve composite plugs of adequate strength they had to be of a significant

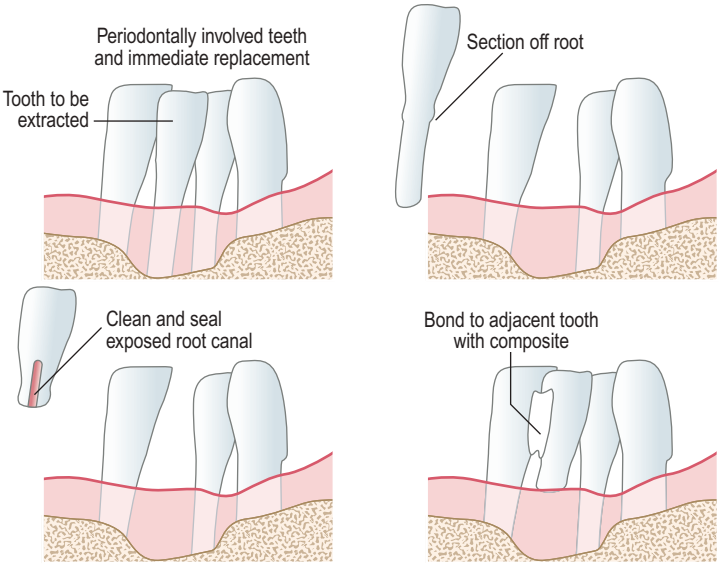


Figure 18.2 Diagrammatic representation of the patient seen in Figure 18.3. The teeth are periodontally involved and the lower right central incisor requires extraction. Following extraction the root is resected (top right), the root canal is cleaned from the resected end and the canal restored with composite resin or glass ionomer (bottom left). The tooth is then bonded to the adjacent tooth with composite resin cement (bottom right).



Figure 18.3 Lower central incisor resin retained natural tooth pontics cantilevered off the lateral incisors.

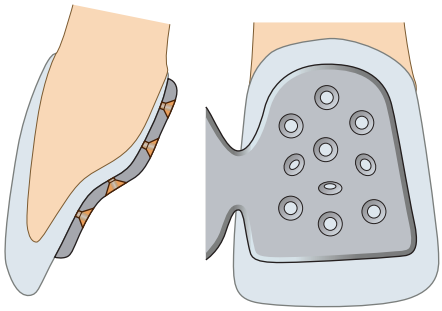


Figure 18.4 Diagrammatic representation of a Rochette bridge. The cross-section shows the countersunk holes and retentive composite 'rivets'.



Figure 18.5 An example of a fixed–fixed Rochette bridge. This bridge had been in situ for over 20 years.

diameter and depth, the latter necessitating a thick wing which caused problems in relation to the occlusion, especially when placed in the upper anterior region. In addition, occlusal stresses were concentrated on a limited number of composite plugs, predisposing to them to fracture and, as such, retention rates were often poor in the long term.

Today, Rochette bridges can still be considered as an immediate provisional tooth replacement when ease of removal is a benefit. For example, if a patient was unable or unwilling to wear a removable prosthesis and an upper central incisor required extraction, its immediate replacement with a cantilever Rochette bridge, using the contralateral central incisor as an abutment, could be considered. The tooth could be extracted, rubber dam applied and the Rochette bridge cemented without any blood contamination from the socket.

Provided the bridge is cemented in place with a *non-adhesive*, dual or chemically cured composite resin luting cement, such as Nexus luting cement (Kerr/SDS), removal is achieved by simply removing the composite from the holes within the wing. It is important to note that if the permanent tooth replacement is to be a minimal preparation (resin-bonded) bridge using the same abutment tooth, the superficial enamel will be impregnated with resin and this will also have to be removed, otherwise the subsequent bond for the permanent bridge will be compromised. Therefore, this technique is preferred when either a different tooth is to be used as an abutment for the definitive bridge or if the final replacement is an implant which is not reliant on a tooth for support and retention.

Micromechanical retention and solid wing retainers

Rochette bridges suffered from a number of problems, namely potential degradation of the exposed composite (wear) and areas of stress concentration within the countersunk holes and composite rivets. In an attempt to retain a solid wing with reduced thickness and provide an improved retention for minimal preparation (resin retained) bridges Thompson and Livaditis in 1982 described a method of electrolytically etching the fit surface of non-precious alloys. The acid, its concentration and etching time were different for different alloys. In each case, however, an etch pattern was produced that allowed micromechanical retention of the bridge when a chemically cured non-adhesive resin was used for cementation. This bridge became known as the 'Maryland bridge'. The Maryland bridge technique required acid baths in the laboratory, and was a technique-sensitive procedure. Attempts were made to use chairside acid etch gels for the metal frame; however, the introduction of adhesive resin cements in the late 1980s superseded their further development.

Adhesive resin luting cements

Unlike resin luting cements, adhesive resin luting cements have the ability to bond to metal in addition to etched



Figure 18.6 Resin retained bridge. Fit surface of wing before (left) and after sandblasting (right).

tooth tissue. Such cements include those that contain 4-META (4-methacryloxyethyl trimellitate anhydride) or phosphate groups (e.g. Panavia) which allow a chemical bond to form with metal. Superior bonding of these cements occurs with surface-abraded non-precious metal alloys such as cobalt chromium or nickel chromium. Whilst the latter allows a more rigid casting to be formed, there are problems associated with allergies (to nickel) for laboratory workers, dentists and patients. Precious metal alloys have therefore been used despite earlier reports of inferior bond strengths to precious metals.

To improve the bond strength of adhesive resin luting cements to precious metal alloys, various surface treatments have been used on the fit surface of the retainer. These include sandblasting with 50 μm alumina, heat treatment and oxide formation, silicate coating, tin plating or application of a metal primer. It is generally accepted that roughening the fit surface by sandblasting is the most convenient and reliable surface treatment, providing bond strengths comparable to the other methods when appropriate adhesive resin luting cements are used (Figure 18.6).

PRINCIPLES OF TOOTH PREPARATION AND BRIDGE DESIGN – ANTERIOR TEETH

The retention achieved for resin retained bridges will be dependent on the quantity and quality of the enamel to which the retainer(s) can be bonded. It is important to maximize the surface area of tooth tissue covered whilst not compromising the aesthetics. Although it would seem logical to use more than one wing as a retainer to increase the total surface area, it soon became apparent that when fixed–fixed resin-bonded bridge designs were used, one retainer commonly debonded.



Figure 18.7 A three unit fixed–fixed resin retained bridge was made to replace the upper right lateral incisor. The wing had debonded off the canine and has now been cut off. The bridge has now been successful as a cantilever bridge.

For example, the patient in Figure 18.7 has a missing upper right lateral incisor. This was originally replaced with a fixed–fixed minimal preparation (resin-bonded) bridge using both the upper right canine and central incisor as abutment teeth. It can be seen that the surface area available for bonding to on the palatal surface of the canine tooth is less than that of the central incisor tooth. Thus the retention is also less. This, together with differences in the direction of the occlusal forces on each abutment tooth, led to the canine retainer debonding. The wing on the canine tooth has been removed and the bridge now functions as a cantilever bridge off the upper right central incisor tooth. A recent literature review comparing fixed–fixed three unit resin retained bridges with two unit cantilever designs has shown that the longevity of the latter is better when used in similar clinical situations (van Dalen et al., 2004).

There is some debate as to how much tooth preparation is required for minimal preparation (resin-bonded) bridges. A cingulum rest is important to resist potential shear stresses within the cement lute which would result from occlusal forces (Figure 18.8). The cingulum rest also helps as a locating device for cementation as it can be difficult to determine the correct seating of the wing onto a smooth lingual/palatal tooth surface. Even with such a rest seat it is often difficult to position the bridge during try-in and cementation. To facilitate this, some clinicians prescribe a small incisal extension or rest from the wing which can be cut off once cemented

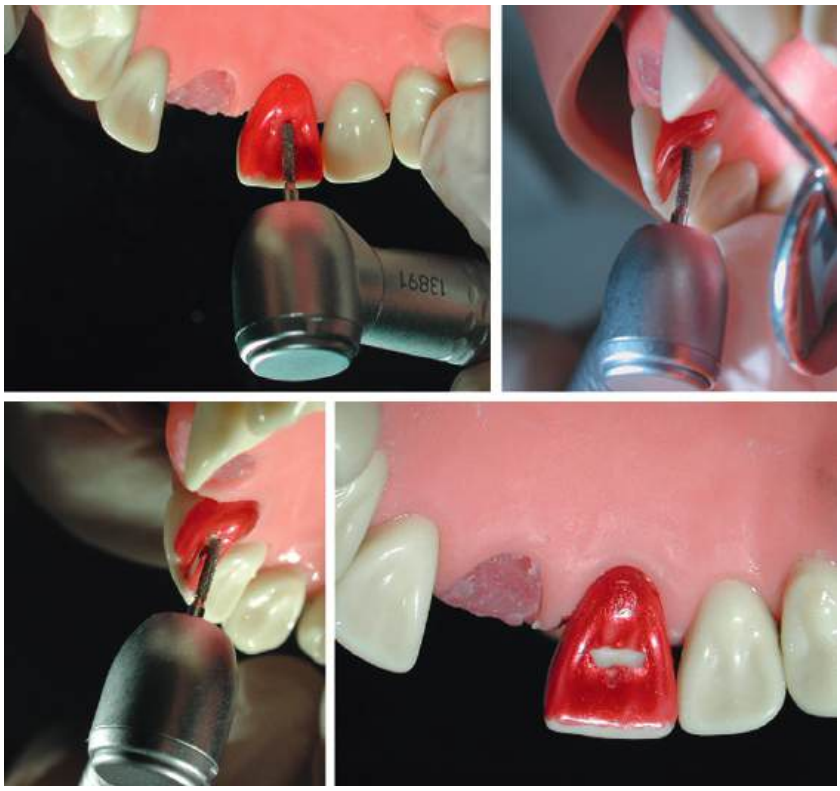


Figure 18.8 Cingulum rest preparation for a cantilever resin retained bridge.

(Figure 18.9). This is done by sectioning the rest along the red dotted line as seen in Figure 18.9 and using an excavator to lever it off whilst protecting the airway with a gauze square or Spontex sponge and high volume suction. The rest is easy to remove as the incisal and labial aspect of the tooth is not etched, the incisal rest fit surface is not sandblasted and no luting cement need be applied to it. Whilst such rests help the operator in positioning the bridge for cementation, they hinder patients when trying to assess the appearance. Once the rest has been removed the metal wing can be polished (Figure 18.10).

Some clinicians also like to cut a cervical chamfer whilst others do not. Those that support this view argue that it aids in checking the seating of the casting onto the tooth and avoids an overhanging cervical margin (Figure 18.11). However, in the area that the chamfer is cut, the enamel cap is very thin and dentine can be exposed. This might lead to sensitivity until the bridge is cemented and the bond of the luting cement to dentine is not as strong as it is to etched enamel. With the seating rests described and a cleansable rounded cervical margin to the wing, the claimed advantages of preparing the tooth cervically are relatively minor.

Occlusal clearance

For patients with a complete tooth-to-tooth overbite, placement of a resin retained bridge in the upper anterior region will lead to premature occlusal contact of the lower incisors with the wing retainer. There are three solutions to this:

- Adjust the opposing tooth when the minimal preparation (resin-bonded) bridge is fitted. This compromises the opposing tooth and may affect the appearance.
- Create occlusal clearance at the time of tooth preparation. However, overeruption of the opposing teeth might take place whilst the bridge is being made in the laboratory, losing the valuable space created. Temporization of the palatal surface of the upper anterior teeth is not possible unless an adhesive material is used. This poses difficulty when trying to remove it accurately at the subsequent fit appointment as it is important not to leave any material behind or inadvertently remove some of the tooth surface which would result in a poor fit of the bridge. If composite is placed, impregnation of the etched enamel with resin also compromises the



Figure 18.9 Incisal extensions (rests) from the metal wings aid the technician and clinician in accurately seating and cementing the minimal preparation (resin retained) bridge; however, assessment of appearance is difficult (bottom right). The rests can be cut along the dotted line (bottom left) and removed after cementation.



Figure 18.10 A similarly designed bridge to that in Figure 18.9 with the incisal rests removed and polished.

subsequent bond of the bridge. Placement of composite onto the incisal edge of the opposing tooth overcomes these problems and will maintain the occlusal clearance. This can then simply be removed once the bridge is cemented.

- The bridge can be made and cemented without occlusal clearance, provided the patient knows at the treatment planning stage that the bridge will feel 'high on the bite' at the fit appointment. This will then not appear to the patient as a lack of skill or judgement. The bridge will then act as a Dahl appliance (Ricketts & Smith, 1993a,b), causing intrusion of the opposing teeth and allowing the remaining separated teeth to overerupt (see Chapter 16). Patients adapt quickly to this change and a normal occlusion should be achieved within 2–3 months.

Special considerations – missing lateral incisors

Hypodontia (up to six congenitally missing teeth) is common, third molar teeth are most often affected and up to 2% of individuals suffer from one or both missing lateral incisor teeth. In this situation a decision has to be made whether to create space for the upper lateral incisors with orthodontic treatment followed by prosthodontic replacement, which usually results in optimum aesthetics, or to close the space and modify the shape of the canine teeth. The latter often results in a compromised appearance, but the patient has no prosthodontic replacement to be maintained throughout life (see Figure 17.18, Chapter 17).

If space is created for a prosthodontic lateral incisor tooth and a cantilever minimal preparation (resin retained) bridge is planned, either the central incisor or the canine tooth can be used as an abutment. The central incisor benefits from having a larger surface area to bond to than the canine, but often the central incisors are too translucent and unsightly grey shine through from the metal wing can occur (Figure 18.12). In these situations the canine teeth have to be used, even though this might affect guidance in lateral excursion.

The mesial surface of the canine tooth is often markedly convex, with its bulbosity close to the gingival margin. This poses two problems: (1) the connector between the wing and the pontic is too narrow for adequate strength and rigidity (Figures 18.13 and 18.14); and (2) it is too close to the gingival papilla to allow for oral hygiene procedures and best appearance. Preparation of a guide plane on the mesial surface of the canine significantly increases the dimensions of the connector (Figure 18.13) and allows it to be moved further away from the gingiva,

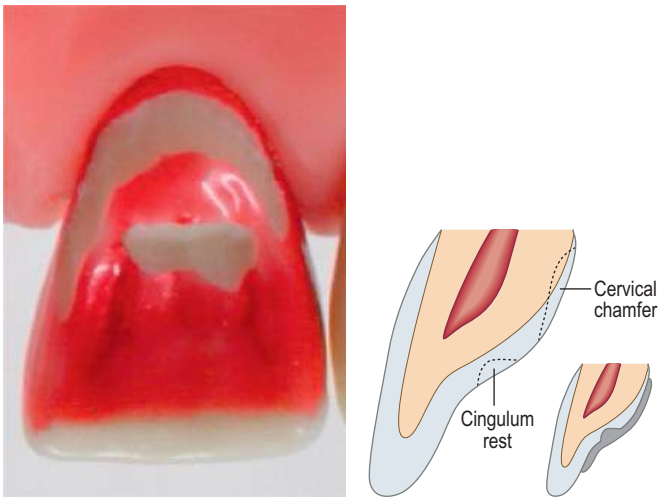


Figure 18.11 Cervical chamfer preparation for resin retained bridge (left). The diagrammatic cross-section through the tooth shows that this can often expose dentine due to the tapered enamel cap (top right). If no cervical preparation is carried out, finishing the wing with a polished and rounded margin should facilitate oral hygiene procedures (right).



Figure 18.12 Resin retained bridges using the central incisors as abutment teeth has led to shadowing as the incisors are very translucent.

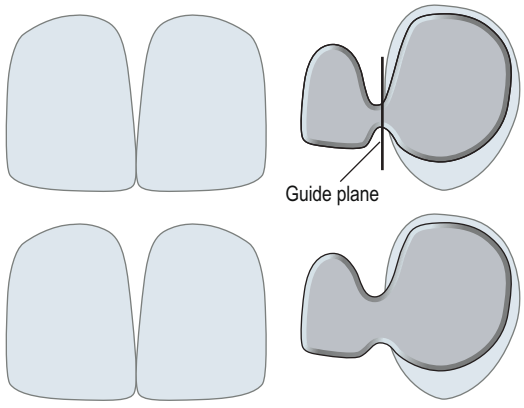


Figure 18.13 Diagrammatic representation of a cantilever resin retained bridge to replace a missing lateral incisor using the canine as an abutment. If no preparation is carried out, the connector is often too thin (top); cutting a small mesial guide plane significantly increases its dimension (bottom).

overcoming these problems. This principle can be applied in a number of clinical situations and allows a definite path of insertion.

Preventing orthodontic relapse

When patients have undergone orthodontic treatment to create space for a prosthodontic replacement tooth (or teeth) there is still the potential for the orthodontic treatment to relapse (Figure 18.15). It is important that at the fit appointment once the bridge has been cemented an impression is taken for a vacuum-formed orthodontic retainer to be made which can be worn over the teeth and bridge at night. Orthodontic retention should be regarded as long term and continued throughout life. Some orthodontic movements are more susceptible to relapse than others and closure of a midline diastema is an example of a particularly unstable procedure. It is possible to use resin retained bridges as orthodontic retainers in this instance. The patients seen in Figures 18.9 and 18.10 have had the wing retainers linked together to prevent the corrected midline diastema from relapsing. Care needs to be taken to ensure that the patient can use super floss beneath the connector between the wings and good follow-up is required to confirm that one retainer does not debond.



Figure 18.14 The connector between the upper left lateral incisor wing retainer and central pontic is too thin and has fractured. This extensive bridge design was probably chosen as the lateral incisors alone are unlikely to provide adequate retention for two cantilever bridges. Implant retained prostheses would have had a better prognosis.



Figure 18.15 This patient has been provided with two cantilever resin retained bridges following orthodontic treatment to create space for the lateral incisors. No orthodontic retainer was provided and as such there has been orthodontic relapse leading to opening of a midline diastema and slippage of the contact point with the pontic moving labially.

PRINCIPLES OF TOOTH PREPARATION AND BRIDGE DESIGN – POSTERIOR TEETH

The same principles of tooth preparation apply to posterior minimal preparation (resin retained) bridges. Like anterior bridges there may be little benefit in preparing a cervical chamfer, but occlusal rest seats are important to resist displacement from occlusal loads and stresses placed on the pontic that would lead to cohesive and adhesive failure at the cement lute interfaces. As occlusal loads are likely to be greater posteriorly, it is important to try to achieve a 180° wrap-around to the wing retainer. Posterior teeth are also bulbous and preparation of a proximal and lingual guide plane increases the tooth surface area above the survey line to which the bridge can be bonded (Figure 18.16).

It is possible to improve the surface area of the tooth to which the minimal preparation (resin retained) bridges can be cemented by covering as much of the occlusal surface as the occlusion will allow. To assist in the preparation and treatment planning, articulated study casts are invaluable (Figure 18.17). It is clear from the articulated

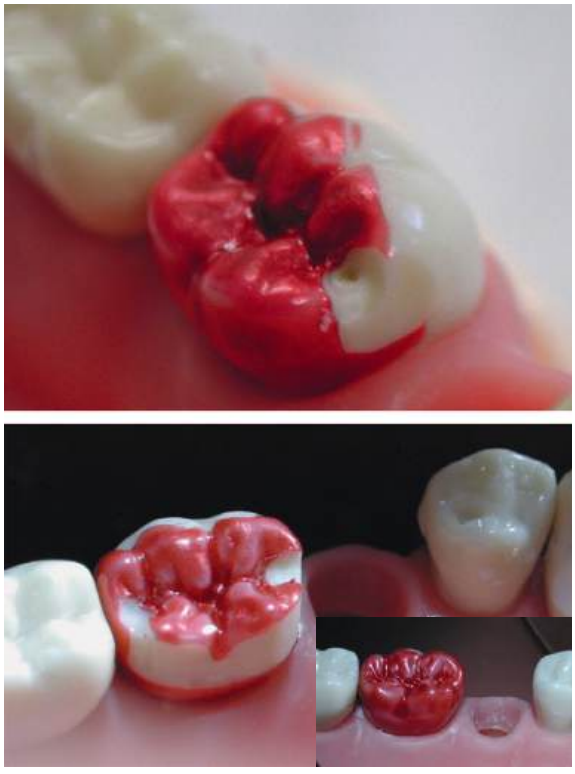


Figure 18.16 Posterior resin retained bridge preparation, showing rest seats mesially and distally and a mesial and lingual guide plane to increase the surface area for bonding.

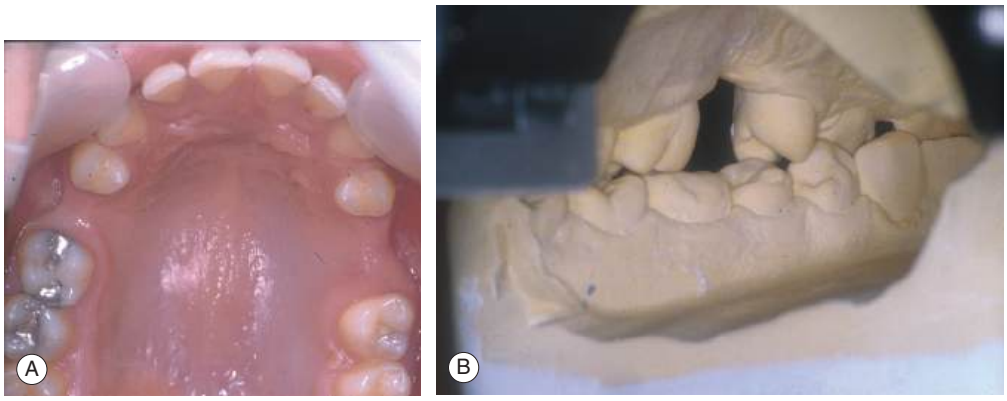


Figure 18.17 Patient with hypodontia requiring prosthetic tooth replacement (A). Articulated study casts show that there is sufficient occlusal clearance over the palatal cusps when in intercuspal position for metal coverage which can be incorporated into the resin retained bridge design (B).



Figure 18.18 Fixed-movable resin retained bridge for the patient seen in Figure 17. Covering the palatal cusps has provided excellent retention and has brought the teeth into function.

casts that there is no occlusal contact between the palatal cusps of the potential abutment teeth in the upper arch and the opposing teeth in the intercuspal position. Extending the wing retainer over the cusp tips would significantly improve retention and support against occlusal forces. An additional advantage is that it also brings the abutment teeth into functional occlusion (Figure 18.18). In situations where appearance is not an issue, buccal wing retainers can also be considered.

Abutment teeth can have existing small to moderate sized intracoronal restorations present which can be incorporated into the design of the minimal preparation (resin retained) bridge (Figure 18.19). This will give a degree of conventional mechanical retention to the retainer. Most intracoronal cavities for direct restorations may contain undercuts due to the natural spread of caries and cavity preparation, but indirect preparations need to be divergent occlusally to allow insertion and removal of the restoration. Thus, when the existing restoration is removed, these undercuts have to be eliminated, either by removing more tooth tissue with a tapered bur or by blocking out the undercut with (ideally) an adhesive restorative material. The former is usually overly destructive of tooth tissue, so the latter is preferred, but can be difficult to perform accurately (Figure 18.20). An alternative approach is to restore the entire cavity with the adhesive material and then cut an inlay preparation within it, ensuring all preparation margins are onto tooth tissue (Figure 18.21). For deeper cavities it is wise to ensure that the occlusal aspect of the inlay is only about 3 mm or so deep. This will give adequate retention, but should allow easy access to the pulp chamber through the metal framework if root canal treatment was ever required.

For longer span resin-bonded bridges which have abutment teeth of different root surface areas there is a problem if one tooth potentially has more physiological movement than the other, with the result that debonding occurs at the retainer with least retention (usually at the smaller tooth with the greater potential for movement). To overcome this problem, consider the patient in Figures 18.17 and 18.18. On the left-hand side the edentulous space can be restored with either two separate cantilever bridges, one pontic retained off the premolar tooth and one off the molar tooth, or by using a fixed–movable bridge design which allows some differential tooth movement of the two abutment teeth by virtue of the movable joint.

The movable joints incorporated into the castings usually have a factory-made precision fit and little movement other than along the long axis of the joint occurs. If the abutment tooth is restored it is possible to simply cut a rest seat within the restoration. This will give support to the pontic at its furthest point from the retainer but little bracing in excursive movements. For this type of bridge design careful attention needs to be paid to the occlusion to avoid heavy lateral loads on the pontic.



Figure 18.19 The mesio-occlusal restoration has been removed from the upper first molar tooth and the cavity has been incorporated into the resin retained bridge design giving superior retention.

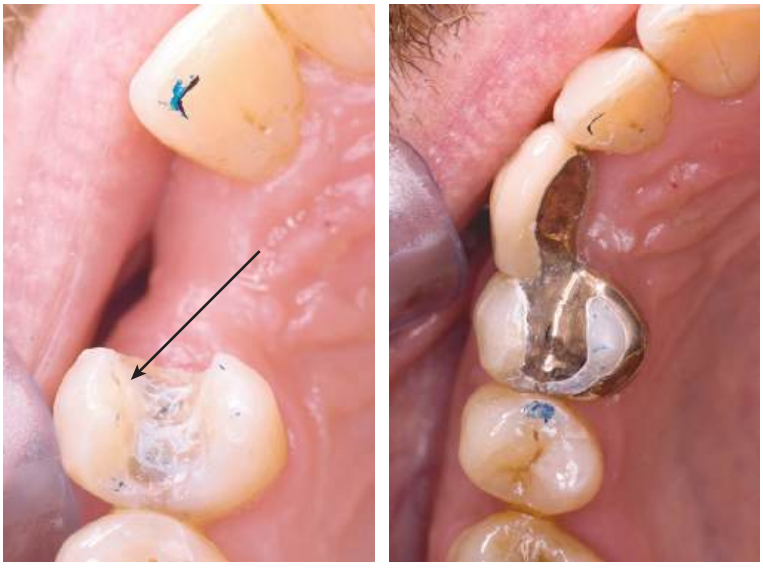


Figure 18.20 Undercut on the buccal aspect of the cavity has been blocked out with composite (arrowed), producing a divergent inlay preparation. Care has been taken in this situation to avoid occlusal contacts on the canine pontic in lateral excursion.

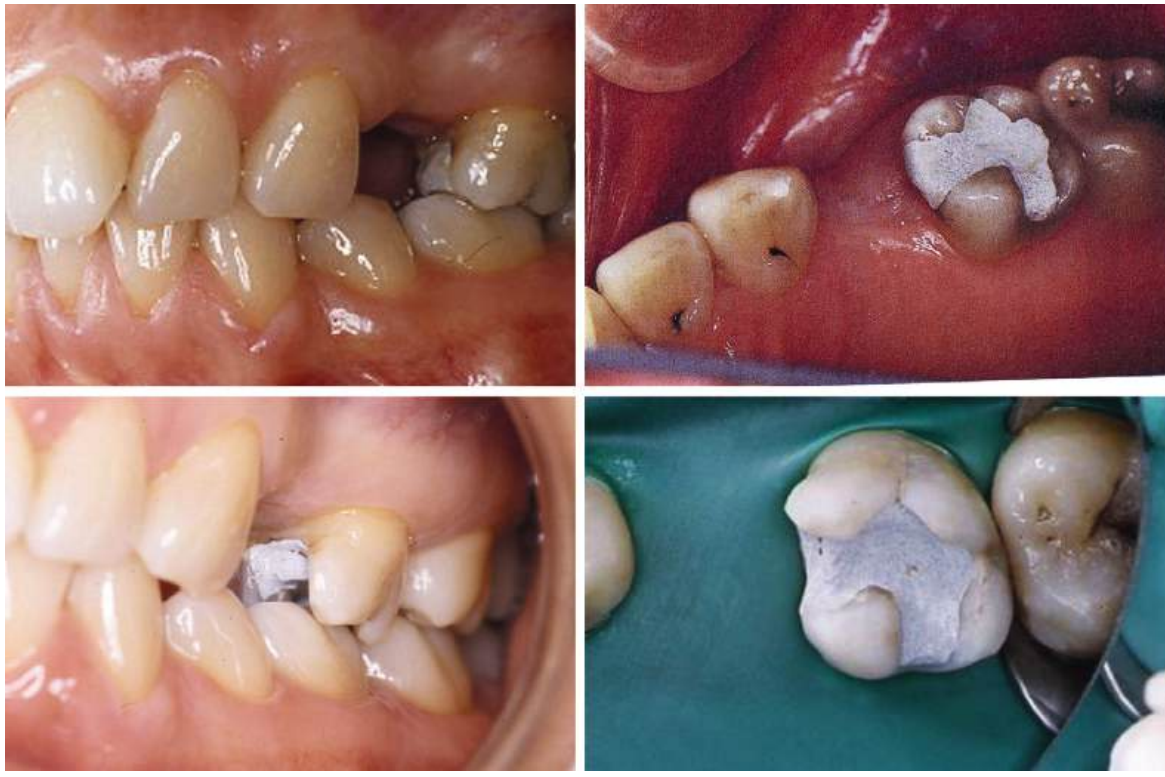


Figure 18.21A The first molar tooth has been restored with a silver re-enforced glass ionomer cement and a classic inlay preparation has been cut leaving the cement to block out the undercuts. The cement at the floor of the box still needs to be removed to finish all margins onto tooth tissue. (bottom left).



Figure 18.21B The minimal preparation (resin retained) bridge cemented.



Figure 18.22 The resin retained bridge in the lower right quadrant has an inlay retainer on the second molar tooth and a rest seat has been cut in a disto-occlusal composite in the second premolar tooth to gain support at the mesial end of the pontic.

Such bridge designs are useful posteriorly where there is canine guidance in lateral excursion which results in posterior disclusion and a posterior protected occlusion (Figure 18.22).

It is important that rest seats are only used within restorations and not natural tooth tissue, as – unlike removable cobalt chromium prostheses – the restoration cannot be removed to clean beneath the rest and there is a risk that caries will occur beneath it. The rest seat is not cemented to the restoration, but sits passively on it. As there should be no obvious movement of the rest seat, wear of the restorative material should not be an issue.

SUMMARY

The introduction of adhesive resin luting cements has completely changed the way in which dentists manage the replacement of missing teeth in a dentition which is minimally restored. If care is taken in optimizing the retention of resin retained bridges through the principles outlined in this chapter and the use of a meticulous cementation technique, long-term success can be guaranteed.

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